EXECUTIVE SUMMARY

Regulated Operational Workflow Performance Analysis

Objective

Evaluate workflow performance within a regulated, high-volume service environment to identify throughput bottlenecks, quantify service-level compliance, assess safety exposure, and recommend governance-aligned operational improvements.

Scope

- 6,000 workflow encounters
- 30-day performance window
- Multi-stage process: Registration → Triage → Queue → Service
- Segmented by appointment type (Walk-in vs Scheduled) and risk level (Low / Medium / High)

Defined performance thresholds:

- **SLA:** Patients seen within 20 minutes
- **Safety Threshold:** High-risk cases should not exceed 10-minute queue wait

Key Findings

1. Structural Bottleneck Identified

Queue wait time averaged **25.8 minutes**, representing ~44% of total cycle time (58.5 minutes). Registration and triage durations remained stable (~7 minutes each), confirming that consultation capacity during peak periods is the primary throughput constraint.

2. Service-Level Compliance Is Critically Low

Only **33.9%** of patients met the 20-minute SLA target. This indicates systemic underperformance relative to acceptable service standards rather than isolated variability.

3. Demand Mix Drives Performance Instability

Walk-in patients demonstrated significantly lower SLA compliance (**26.8%**) compared to scheduled patients (**43.9%**). Uncontrolled walk-in volume materially degrades throughput stability during peak windows.

4. Safety Exposure Persists Under Peak Strain

Although high-risk patients are prioritized (average wait ~12 minutes vs ~29 minutes for low-risk), **66.3%** of high-risk cases exceed the 10-minute safety threshold during peak hours, indicating prioritization erosion under congestion.

5. Predictable Peak Failure Window

Performance degradation is concentrated between **9 AM–12 PM**, where average wait times exceed 29 minutes and SLA compliance falls below 25%. This congestion window is structurally predictable and operationally addressable.

Recommendations

1. Peak-Hour Walk-In Load Management

Implement triage-based gating or deferral of non-urgent walk-ins during 9–12 PM to stabilize demand and protect SLA performance.

2. High-Risk Fast-Track Protocol

Establish a dedicated consultation lane during peak hours to reduce safety threshold breaches.

3. Governance Monitoring Framework

Track SLA compliance, high-risk breach rate, hourly performance heatmap, and walk-in ratio daily. Trigger operational review if peak-hour SLA falls below 40% for three consecutive days.

Strategic Impact

This analysis transforms ambiguous operational complaints into measurable performance diagnostics. By structuring KPIs around service-level compliance, bottleneck isolation, and safety exposure, leadership gains actionable visibility to improve throughput, strengthen governance, and reduce regulated risk.